

The Balanced Pair in Excitable Tissue:

What a Membrane Reading Determines, and the Balanced Cortex

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Abstract

Excitable tissue is read through pairs: the open and closed states of a gate, the inward and outward charges of a spike, the excitatory and inhibitory inputs of a cortical network. We prove that in a balanced network of excitatory and inhibitory neurons the residual net input and the distance of the population rates from the balance solution are one object, $\mu = \sqrt{K} A(\nu - \nu_{\text{bal}})$, where K is the number of connections per neuron and A the matrix of the balance equations. In simulations of 2×10^4 binary neurons with $K = 100$ to 1600 this identity reproduces the measured rates to 2.67×10^{-3} . The excitatory and inhibitory inputs grow as $\sqrt{K}$ per unit rate (log-log slopes 0.4994–0.5004), the net input stays of order one, and fractions 0.9763 (E) and 0.9954 (I) of the excitatory drive cancel at $K = 1600$. The rates approach balance slowly, 40.6% (E) and 22.2% (I) below it at $K = 1600$, exactly as the identity requires: $\det A = 0.2$ amplifies the residual, and mean-field rates come within 10% of balance only for $K > 40,054$ (E) and $K > 8,897$ (I).

The network is the top of a ladder that starts at the single gate. The two-state gate $p = 1/(1 + e^{-z(V - V_{1/2})})$ is balanced at its half-point, where its occupancy is $\frac{1}{2}$, its entropy one bit and its variance largest. The Hodgkin–Huxley half-points are -40.02 (m), -62.31 (h) and -53.41 mV (n), and n^4 reaches half its maximum 34.5 mV above the half-point of its gate. Mean currents determine only the product Ni of channel number and unitary current, and the variance separates the two; the precision of the channel count improves steadily with the largest open probability reached, with no threshold at $\frac{1}{2}$: it goes from no reliable estimate at 0.19 to a factor of two at 0.5 and $\pm 12\%$ at 0.87. Voltage-dependent block is a two-state gate whose two parameters separate affinity from electrical distance; the magnesium block of NMDA receptors has its half-point at -20.5 mV at 1 mM, moving 11.2 mV per doubling. We derive the spike threshold of exponential-onset sodium activation in closed form, $\theta = E_{\text{Na}} - k_a[1 - W_{-1}(-(g_L/g_{\text{Na}})e^{1-(E_{\text{Na}}-V_a)/k_a})]$, which reduces the error of the simplest threshold equation by a factor of 1.3 to 33, and show that a threshold reading fixes V_a and g_{Na} only through $g_{\text{Na}}e^{-V_a/k_a}$. The inward and outward charges of a spike cancel exactly. The sodium conductance m^3h is a triad with a fourth: three activating particles, like the channel’s three fast voltage sensors, and one inactivating particle, like the domain IV sensor. Its spike (amplitude 104.06 mV) carries 13.36 times the minimum sodium charge, and among exponents 1 to 4 only m^3h gives both a stable rest and a spike. The sodium–potassium pump is the unequal pair 3:2; it moves one elementary charge per cycle and stalls at -205 mV. The proofs use the two-state law, the binomial variance, the Lambert function and the balance equations; the numbers come from the Hodgkin–Huxley equations integrated to relative tolerance 10^{-9} and from direct simulation of the network.

1 Introduction

Hodgkin and Huxley described the squid giant axon by voltage-dependent gates: three activating particles m and one inactivating particle h for the sodium conductance, and four particles n for the potassium conductance [1]. Fluctuation analysis added a second reading of the same channels: the variance of a current carried by N channels is a parabola in its mean, from which

Neher and Stevens and Sigworth obtained the number of channels and their unitary current [2, 3, 4]. Woodhull described voltage-dependent block by a charged blocker that binds partway through the membrane field [5]; the magnesium block of NMDA receptors [6, 7] is its best-known instance, with the empirical form of Jahr and Stevens [8]. Fourcaud-Trocmé et al. showed that spike initiation is captured by an exponential onset of sodium activation [9], and Platkiewicz and Brette derived a threshold equation for it [10]. Fluorescence recordings placed the fast activation of the sodium channel in the voltage sensors of domains I–III and its inactivation in the sensor of domain IV [11, 12]. At the scale of the cortex, van Vreeswijk and Sompolinsky showed that a network whose excitatory and inhibitory inputs are each of order $\sqrt{K}$ settles into a balanced state in which they cancel to order one [13, 14].

Each of these readings is a reading of a pair. The two-state gate has log-odds $-z(V - V_{1/2})$, which takes the values s and $-s$ at $V_{1/2} \mp s/z$, with the half-point as its zero; the sum of independent gates keeps the half-point; the connectivity row $\{+1, 0, -J_X\}/\sqrt{K}$ of the balanced network has excitatory and inhibitory members that cancel to order one; and the inactivating particle of m^3h closes what the other three open. The classical questions (how many channels there are, how large a single-channel current is, where a gate activates, where a spike starts, how far a network is from balance) are therefore questions of identifiability: which parameters of the hidden pairs a reading determines, and which it confounds.

Companion papers. The two-state law used here, $p = 1/(1 + e^\varepsilon)$ with ε the log-odds of the closed member, is the law with which [16] reads atomic levels and gas sensors. There ε is an energy difference in units of $k_B T$ (with a degeneracy term) and the half-point is a temperature. Here $\varepsilon = -z(V - V_{1/2})$ and the half-point is a voltage; for a gate moving an effective gating charge z_g through the membrane field, $\varepsilon = -z_g e(V - V_{1/2})/k_B T$ is again a free-energy difference in units of $k_B T$ [17]. The ternary rows of the cortical connectivity are weight vectors of the kind used in the ternary-weight networks of [18], with the magnitude split between $+1$ and $-J_X$. The synthesis paper [15] places these results beside those of the other fields.

1.1 Main results

The balanced cortex. In the network of van Vreeswijk and Sompolinsky (Section 7), N_E excitatory and N_I inhibitory binary neurons are connected at random: a neuron of population $X \in \{E, I\}$ receives on average K connections from each population, with weight $+1/\sqrt{K}$ from an excitatory neuron and $-J_X/\sqrt{K}$ from an inhibitory one, together with an external input $\sqrt{K} b_X \nu_0$. Write $\nu = (\nu_E, \nu_I)^\top$ for the population rates, $\mu = (\mu_E, \mu_I)^\top$ for the population-mean net inputs, and

$$A = \begin{pmatrix} 1 & -J_E \\ 1 & -J_I \end{pmatrix}, \quad b = \begin{pmatrix} b_E \\ b_I \end{pmatrix}.$$

Theorem 1 (the zero and the residual). *Let $\nu_{\text{bal}} = -\nu_0 A^{-1} b$, that is*

$$\nu_E^{\text{bal}} = \nu_0 \frac{J_I b_E - J_E b_I}{J_E - J_I}, \quad \nu_I^{\text{bal}} = \nu_0 \frac{b_E - b_I}{J_E - J_I}.$$

(a) *Both are positive if and only if $b_E/b_I > J_E/J_I > 1$ or $b_E/b_I < J_E/J_I < 1$.*

(b) *Up to terms of order $\sqrt{K}/N_E$ from the excluded self-connections, the residual net input and the rate error are one object:*

$$\mu = \sqrt{K} A (\nu - \nu_{\text{bal}}), \quad \nu - \nu_{\text{bal}} = \frac{A^{-1} \mu}{\sqrt{K}}.$$

(c) *Hence rates that stay bounded with $\mu = O(1)$ converge to ν_{bal} with error $O(1/\sqrt{K})$, amplified by A^{-1} , whose entries scale as $1/\det A = 1/(J_E - J_I)$.*

We simulate the network with $N_E = N_I = 10^4$, $J_E = 2$, $J_I = 1.8$, $b_E = 1$, $b_I = 0.8$ and $\nu_0 = 0.1$, for which $\nu_{\text{bal}} = (0.1, 0.1)$ and $\det A = 0.2$, at $K = 100$ to 1600 (Section 7). Inserting the measured net inputs into Theorem 1(b) reproduces the measured rates to within 2.67×10^{-3} at every K . The excitatory and inhibitory inputs grow as $\sqrt{K}$ per unit rate, with log-log slopes 0.4994–0.5004; their raw slopes, 1.0962 (excitation) and 0.6774 (inhibition) onto the E population, are $\frac{1}{2}$ plus the slopes of the still-rising rates. The net input stays of order one (log-log slopes -0.0523 and 0.0234), and at $K = 1600$ fractions 0.9763 (E) and 0.9954 (I) of the excitatory drive cancel. The rates approach balance slowly, 40.6% (E) and 22.2% (I) below it at $K = 1600$, exactly as Theorem 1 requires: A^{-1} multiplies the order-one residual by up to 10, and in mean field the rates come within 10% of balance only for $K > 40,054$ (E) and $K > 8,897$ (I). The mean net input lies 1.66 (E) and 1.55 (I) standard deviations of its fluctuation below threshold, so the neurons fire on fluctuations, irregularly and asynchronously.

The gate. The network is the top of a ladder that starts at the single gate. A gate x with opening rate $\alpha(V)$ and closing rate $\beta(V)$ relaxes to $x_\infty = \alpha/(\alpha + \beta)$ with time constant $\tau = 1/(\alpha + \beta)$ [1]. Its half-point $V_{1/2}$ is where the two members exchange at equal rates, $\alpha(V_{1/2}) = \beta(V_{1/2})$, and $\mathcal{H}(x) = -x \log_2 x - (1 - x) \log_2 (1 - x)$ is the entropy of a single gate's state.

Theorem 2 (the gate at its half-point). *Let $\varepsilon(V) = \ln(\beta/\alpha)$ be the log-odds of the closed member.*

- (a) $x_\infty = 1/(1 + e^\varepsilon)$ and $x_\infty - \frac{1}{2} = -\frac{1}{2} \tanh(\varepsilon/2)$.
- (b) At $V_{1/2}$: $\varepsilon = 0$, $x_\infty = \frac{1}{2}$, $\tau = 1/(2\alpha(V_{1/2}))$, the entropy $\mathcal{H}(x_\infty)$ takes its maximum, one bit, and the variance $x_\infty(1 - x_\infty)$ of the gate's state takes its maximum, $\frac{1}{4}$.
- (c) $\text{logit } x_\infty = \ln(x_\infty/(1 - x_\infty)) = \ln(\alpha/\beta)$. The gate has the Boltzmann form $p(V) = 1/(1 + e^{-z(V - V_{1/2})})$ for all V if and only if $\ln(\alpha/\beta)$ is linear in V , and then $z = 1/k$ with $k = [\frac{d}{dV} \ln(\alpha/\beta)]^{-1}$. For a general gate this k , evaluated at $V_{1/2}$, is the local slope factor.
- (d) For a Boltzmann gate, $p(V_{1/2} + \Delta) + p(V_{1/2} - \Delta) = 1$ for every Δ , and the occupancies $\frac{1}{3}$ and $\frac{2}{3}$ lie at $V_{1/2} \mp k \ln 2$, where the log-odds are $\pm \ln 2$.

For the Hodgkin–Huxley gates the half-points are -40.02 (m), -62.31 (h) and -53.41 mV (n) (Table 1). A product of gates is not a pair: n_∞^4 crosses $\frac{1}{2}$ at -18.88 mV, 34.5 mV above the half-point of n .

Channel number and unitary current. Let N independent channels each carry unitary current i when open, with open probability p at a fixed voltage, and let a recording add independent background noise of variance σ_b^2 .

Theorem 3 (what the mean and the variance determine). (a) *The mean current $I = Nip$ is invariant under $(N, i) \mapsto (\lambda N, i/\lambda)$ for every $\lambda > 0$: no set of mean measurements determines N or i separately.*

(b) *The variance satisfies $\sigma^2 = \sigma_b^2 + iI - I^2/N$ [3, 2], and measurements at three or more distinct values of p determine (σ_b^2, i, N) .*

(c) *The curvature term I^2/N is a fraction p of the linear term iI . The variance is largest, $\sigma_b^2 + Ni^2/4$, at $p = \frac{1}{2}$.*

By (c), when p stays small the parabola is nearly a line and N is weakly determined. In simulated recordings of $N = 2000$ channels the precision of the count improves steadily with the largest open probability p_{max} reached, and there is no threshold at the half-point: at $p_{\text{max}} = 0.19$ there is no reliable estimate, reaching $\frac{1}{2}$ determines N to within about a factor of two (10–90%

range 1462–3296), and reaching 0.87 to about $\pm 12\%$ (Table 2). A blocker is read in the same way: the Woodhull blocker is a Boltzmann gate whose slope and half-point separate electrical distance from affinity (Theorem 7), and the magnesium block of NMDA receptors has its half-point at -20.53 mV at 1 mM, moving 11.18 mV per doubling of the concentration.

The threshold. Consider $C\dot{V} = \Phi(V) = g_L(E_L - V) + g_{\text{Na}}m_\infty(V)(E_{\text{Na}} - V)$, with m_∞ a Boltzmann gate of half-point V_a and slope k_a , and let the threshold θ be the voltage where $\Phi'(\theta) = 0$ [10]. Below V_a , $m_\infty \approx e^{(V-V_a)/k_a}$ and $1 - m_\infty \approx 1$; we call this the exponential approximation.

Theorem 4 (closed-form threshold). *In the exponential approximation:*

(a) *the threshold satisfies the fixed-point equation*

$$\theta = V_a - k_a \ln \left[\frac{g_{\text{Na}}}{g_L} \left(\frac{E_{\text{Na}} - \theta}{k_a} - 1 \right) \right],$$

whose solution below $E_{\text{Na}} - 2k_a$ is

$$\theta = E_{\text{Na}} - k_a \left[1 - W_{-1} \left(-\frac{g_L}{g_{\text{Na}}} e^{1 - (E_{\text{Na}} - V_a)/k_a} \right) \right],$$

with W_{-1} the lower real branch of the Lambert function;

(b) *the sodium current below V_a depends on V_a and g_{Na} only through $g_{\text{Na}}e^{-V_a/k_a}$, and, given k_a , E_{Na} and g_L , the threshold and all subthreshold dynamics determine only $V_a - k_a \ln(g_{\text{Na}}/g_L)$.*

For $V_a = -30$ mV and $k_a = 6$ mV the closed form reduces the error of the simplest threshold equation of [10] by a factor of 1.3 (at $g_{\text{Na}}/g_L = 2$) to 33 (at 14) (Table 4).

The spike, the 3 + 1 channel and the 3:2 pump. For a space-clamped membrane $C\dot{V} = -I_{\text{Na}} - I_{\text{K}} - I_{\text{L}} + I_{\text{stim}}$, let $Q_{\text{Na}} = \int (-I_{\text{Na}}) dt$ be the inward sodium charge and $Q_{\text{K}} = \int I_{\text{K}} dt$, $Q_{\text{L}} = \int I_{\text{L}} dt$, $Q_{\text{stim}} = \int I_{\text{stim}} dt$, and let $\rho = Q_{\text{Na}}/(C \Delta V_{\text{sp}})$ compare the sodium charge of a spike of amplitude ΔV_{sp} with the charge the membrane capacitance needs to reach the peak.

Theorem 5 (the spike as a pair of charges, and the 3+1 channel). (a) *Over any interval whose end-points have equal voltage, $Q_{\text{Na}} = Q_{\text{K}} + Q_{\text{L}} - Q_{\text{stim}}$. Over the rising phase, from rest to the peak, $Q_{\text{Na}} - Q_{\text{K}} - Q_{\text{L}} + Q_{\text{stim}} = C \Delta V_{\text{sp}}$.*

(b) *In the Hodgkin–Huxley model of the squid axon (Section 6) with sodium conductance $g_{\text{Na}}m^q h$ and all other parameters fixed, $q = 3$ is the only exponent in $\{1, 2, 3, 4\}$ for which the membrane has a stable resting state and fires a spike in response to a pulse of $10 \mu\text{A}/\text{cm}^2$ for 1 ms. For $q = 3$ the spike has amplitude 104.063 mV and $\rho = 13.3566$.*

The sodium conductance m^3h is thus a triad with a fourth: three identical activating particles, as the channel has three fast voltage sensors in domains I–III, and one inactivating particle, as the sensor of domain IV drives fast inactivation [11, 12]. The integrated charges of the squid spike satisfy (a) to 1.0×10^{-16} of Q_{Na} , and 87.35% of the sodium charge enters after the peak.

Theorem 6 (the pump as the unequal pair 3:2). *A pump that exports three Na^+ and imports two K^+ per cycle moves one elementary charge outwards per cycle, twice the half-difference of the pair (3,2), whose common part is $\frac{5}{2}$. With reversal potentials E_{Na} , E_{K} and membrane potential V_m , the work against the gradients per cycle is*

$$W_{\text{cyc}} = 3e(E_{\text{Na}} - V_m) + 2e(V_m - E_{\text{K}}) = e(3E_{\text{Na}} - 2E_{\text{K}} - V_m),$$

which decreases by one electron-volt per volt of V_m . If each cycle is driven by the free energy $|\Delta G_{\text{ATP}}|$ of one ATP, the pump runs forward if and only if $V_m > 3E_{\text{Na}} - 2E_{\text{K}} - |\Delta G_{\text{ATP}}|/e$, the stall potential, and reverses its cycle below it.

For $E_{\text{Na}} = +55$ mV, $E_{\text{K}} = -90$ mV and $V_m = -65$ mV, $W_{\text{cyc}} = 0.410$ eV $= 15.3 k_B T$ at 310 K, against 0.290 eV for a 2:2 exchanger that moves no charge; with $|\Delta G_{\text{ATP}}| \approx 0.55$ eV the pump stalls at -205 mV, so it runs forward throughout the physiological range [19, 20].

1.2 Methods of proof

Theorems 2, 3 and 6 and Theorem 4(b) are exact consequences of the two-state law: the logit of a gate, the binomial variance of N gates written in terms of the mean current, the energy balance of one pump cycle, and the scaling symmetry of an exponential activation. Theorem 4(a) reduces the threshold condition to $Ue^U = w$ and selects the branch of the Lambert function by the side of $E_{\text{Na}} - 2k_a$ on which the threshold lies. Theorem 5(a) is the integral of the current equation. Theorem 5(b) is proved by computation: the fixed points of the $m^q h$ membrane are located as roots of the steady-state current and classified by the eigenvalues of the Jacobian, and the response to the pulse is obtained by integrating the four-dimensional system at relative tolerance 10^{-9} , reproduced by an implicit Runge–Kutta integration. Theorem 1 follows from averaging the ternary connectivity over its distribution (Theorem 8) and inverting the 2×2 balance matrix. The network results come from direct simulation of the binary network with asynchronous updates, and the mean-field rates from the equations of [14], solved by Newton-type root finding.

1.3 Organisation and notation

Section 2 proves Theorem 2 and computes the Hodgkin–Huxley half-points; Section 3 proves Theorem 3 and quantifies channel counting; Section 4 treats voltage-dependent block; Section 5 proves Theorem 4; Section 6 proves Theorems 5 and 6; Section 7 proves Theorem 1 and reports the network; Section 8 draws the scales together, and Section 9 lists every input. Voltages are in mV, gating rates in ms^{-1} , conductances in mS/cm^2 , current densities in $\mu\text{A}/\text{cm}^2$ and charges in nC/cm^2 unless stated otherwise. $X, Y \in \{E, I\}$ label the excitatory and inhibitory populations, network inputs are in units of threshold, and time in the network is in units of the excitatory update time τ_E .

2 The gate as a pair

2.1 The half-point

Let the gate x , its rates α, β , its half-point $V_{1/2}$ and the entropy $\mathcal{H}$ be as in Section 1.1. We prove Theorem 2.

Proof. (a) $x_\infty = 1/(1 + \beta/\alpha) = 1/(1 + e^\varepsilon)$, and $x_\infty - \frac{1}{2} = (1 - e^\varepsilon)/(2(1 + e^\varepsilon)) = -\frac{1}{2} \tanh(\varepsilon/2)$. (b) $\alpha = \beta$ gives $\varepsilon = 0$, $x_\infty = \frac{1}{2}$ and $\tau = 1/(2\alpha)$. $\mathcal{H}'(x) = \log_2((1-x)/x)$ vanishes only at $x = \frac{1}{2}$ and $\mathcal{H}$ is strictly concave, so its maximum is $\mathcal{H}(\frac{1}{2}) = 1$. $x(1-x) = \frac{1}{4} - (x - \frac{1}{2})^2 \leq \frac{1}{4}$, with equality only at $\frac{1}{2}$. (c) $x_\infty/(1-x_\infty) = \alpha/\beta$. The Boltzmann form is equivalent to logit $p = z(V - V_{1/2})$, which is linear in V with slope z ; conversely a linear $\ln(\alpha/\beta)$ that vanishes at $V_{1/2}$ equals $(V - V_{1/2})/k$. (d) $1/(1 + e^{-a}) + 1/(1 + e^a) = 1$ for every real a . Next, $p = \frac{1}{3}$ means $e^{-z(V - V_{1/2})} = 2$, that is $V = V_{1/2} - k \ln 2$, and $p = \frac{2}{3}$ means $e^{-z(V - V_{1/2})} = \frac{1}{2}$, that is $V = V_{1/2} + k \ln 2$. $\square$

2.2 The Hodgkin–Huxley gates

We use the rate functions of Hodgkin and Huxley [1], shifted to a resting potential of -65 mV as in [21]. With V in mV and rates in ms^{-1} ,

$$\begin{aligned}\alpha_m &= \frac{0.1(V+40)}{1 - e^{-(V+40)/10}}, & \beta_m &= 4e^{-(V+65)/18}, \\ \alpha_h &= 0.07e^{-(V+65)/20}, & \beta_h &= \frac{1}{1 + e^{-(V+35)/10}}, \\ \alpha_n &= \frac{0.01(V+55)}{1 - e^{-(V+55)/10}}, & \beta_n &= 0.125e^{-(V+65)/80}.\end{aligned}\tag{1}$$

Time constants refer to the original temperature of 6.3°C . The half-points are the roots of $\alpha(V) = \beta(V)$, found by a bracketing root search to better than 10^{-6} mV. The slope k follows by differentiating $\ln(\alpha/\beta)$; for example, $\frac{d}{dV} \ln \alpha_m = \frac{1}{V+40} - \frac{1}{10} \frac{e^{-(V+40)/10}}{1 - e^{-(V+40)/10}}$ and $\frac{d}{dV} \ln \beta_m = -\frac{1}{18}$. The time constant at the half-point is $\tau(V_{1/2}) = 1/(2\alpha(V_{1/2}))$, and the $\frac{1}{3}$, $\frac{2}{3}$ points are the roots of $x_\infty(V) = \frac{1}{3}$, $\frac{2}{3}$ (Table 1).

Table 1: The Hodgkin–Huxley gates at their half-points.

gate	$V_{1/2}$ (mV)	slope $k = [d \logit x_\infty / dV]^{-1}$ (mV)	$\tau(V_{1/2})$ (ms)	$V(\frac{1}{3}), V(\frac{2}{3})$ (mV)	$x_\infty(-65)$
m	-40.02	$+9.47$	0.50	$-46.43, -33.28$	0.053
h	-62.31	-6.95	8.17	$-57.44, -67.09$	0.596
n	-53.41	$+16.35$	4.62	$-63.99, -41.07$	0.318

The sign of k says which member opens with depolarisation (h closes). The $\frac{1}{3}$ and $\frac{2}{3}$ points lie within 1.02 mV of $V_{1/2} \mp k \ln 2$; the largest difference is that of n at $\frac{2}{3}$. Further from the half-point the Hodgkin–Huxley rate functions depart from Boltzmann form. We measure the departure at occupancy x as $|V(x) - V_{1/2} - k \ln(x/(1-x))|$, with $V(x)$ the root of $x_\infty(V) = x$, and evaluate it at the 161 equally spaced occupancies from 0.1 to 0.9. The largest departure is 1.97 mV for m , 0.84 mV for h and 14.22 mV for n (at occupancy 0.9); for n at 0.95 it is 28.8 mV.

Sums are pairs; products are not. The mean current of N independent identical gates is $Nip(V)$, whose normalised form is $p(V)$ itself, with the half-point and slope of one gate (Section 3). A product of gates behaves differently. The steady sodium open fraction $m_\infty^3 h_\infty$, evaluated on a 0.01 mV grid from -100 to 40 mV, peaks at 0.0077 at -31.9 mV (the window open fraction), far below $\frac{1}{2}$. The steady potassium open fraction n_∞^4 crosses $\frac{1}{2}$ at -18.88 mV, where $n_\infty = 2^{-1/4} = 0.841$. This is 34.5 mV above the gate’s own half-point, and the shift is the effect of taking the fourth power. An exact Boltzmann gate with the same $V_{1/2}$ and k would reach $n_\infty = 2^{-1/4}$ at $V_{1/2} + k \ln(2^{-1/4}/(1 - 2^{-1/4})) = -26.20$ mV, with $\ln(2^{-1/4}/(1 - 2^{-1/4})) = 1.6649$; the 7.3 mV difference is the departure of n_∞ itself from Boltzmann form at 0.841 . The half-activation of a conductance measured from macroscopic current is therefore the half-point of the underlying gate only when the conductance is linear in that gate.

3 What the mean and the variance determine

Let N , i , p and σ_b^2 be as in Theorem 3. Because the unitary current depends on the driving force, the theorem is stated at a fixed voltage, where p is varied in time (as in nonstationary fluctuation analysis, for example during a voltage step). Across voltages the same statements

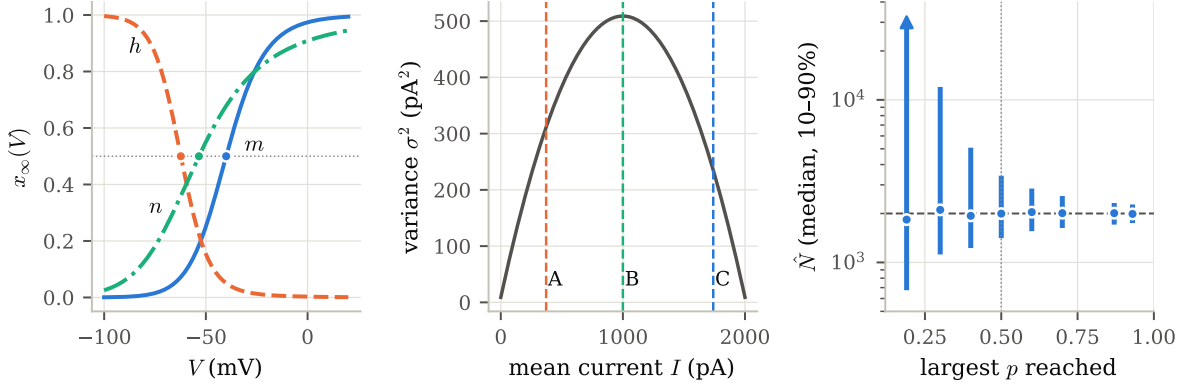


Figure 1: Left: steady-state Hodgkin–Huxley gates with their half-points (dots). Middle: the mean–variance parabola $\sigma^2 = \sigma_b^2 + iI - I^2/N$ ($N = 2000$, $i = 1$ pA, $\sigma_b = 3$ pA); dashed lines mark the largest mean current reached when the largest open probability is 0.19 (A), 0.50 (B) and 0.87 (C). Right: recovered channel number against the largest open probability reached (median and 10–90% range over 400 simulated recordings each; failed fits are counted as $\hat{N} = \infty$, and the arrow marks an infinite upper quantile). Dashed: true N .

hold in conductance units, $\sigma_G^2 = \sigma_b^2/(V - E_{\text{rev}})^2 + \gamma G - G^2/N$, with γ the unitary conductance and G the mean conductance; this is again linear in $(\sigma_b^2, \gamma, 1/N)$.

We prove Theorem 3.

Proof. (a) $(\lambda N)(i/\lambda)p = Nip$. (b) The number of open channels is binomial with mean Np and variance $Np(1-p)$, so the current has variance $\sigma_b^2 + i^2Np(1-p)$. Substituting $p = I/(Ni)$ gives $i^2N \frac{I}{Ni}(1 - \frac{I}{Ni}) = iI - I^2/N$. The right side is a polynomial of degree two in I with coefficients $(\sigma_b^2, i, -1/N)$. Distinct values of p give distinct values of I , and a polynomial of degree at most two is fixed by its values at three distinct points (the 3×3 Vandermonde matrix is invertible). (c) $(I^2/N)/(iI) = I/(Ni) = p$. Also $\frac{d}{dI}(iI - I^2/N) = 0$ at $I = Ni/2$, that is at $p = \frac{1}{2}$, where the variance is $\sigma_b^2 + Ni^2/4$. $\square$

By Theorem 3(c), when p stays small the parabola is nearly a line and N is weakly determined. To quantify this we simulated recordings of $N = 2000$ channels with $i = 1$ pA and $\sigma_b = 3$ pA. Their open probability follows the Boltzmann gate $p(V) = 1/(1 + e^{-(V+40)/9.47})$, which has the half-point and slope of the m gate, rounded. A recording consists of 9 voltages spaced linearly from -80 mV to $V_{\text{max}} = -40 + 9.47 \ln(p_{\text{max}}/(1 - p_{\text{max}}))$ mV, the voltage at which p reaches its largest value p_{max} , with 200 independent sweeps at each voltage. In each sweep the number of open channels is drawn from the binomial distribution with parameters N and $p(V)$. The current is i times that number plus an independent normal deviate of mean 0 and standard deviation σ_b . At each voltage we form the sample mean I and the unbiased sample variance σ^2 of the 200 currents. We then fit $\sigma^2 = \eta_0 + \eta_1 I - \eta_2 I^2$ to the 9 pairs (I, σ^2) by ordinary (unweighted) least squares, giving $\hat{i} = \eta_1$ and $\hat{N} = 1/\eta_2$. A fit fails when $\eta_2 \leq 0$, and such fits are counted as $\hat{N} = \infty$. For each p_{max} we generated 400 independent recordings and report the 10%, 50% and 90% quantiles of $\hat{N}$ (Table 2).

These quantiles are themselves Monte Carlo estimates from 400 recordings. In ten independent repetitions of the whole experiment every quantile for $p_{\text{max}} \geq 0.5$ stayed within 10% of the tabulated value, and the failure fraction stayed at or below 0.5%. For $p_{\text{max}} \leq 0.4$ the 90% quantile lies in a heavy tail and varied by up to 20% ($p_{\text{max}} = 0.4$) and several-fold (0.3); at $p_{\text{max}} = 0.19$ the median varied by up to 25% and the failure fraction between 18% and 27%. None of the conclusions below depends on these variations. The unitary current is recovered in each of the three designs of Fig. 1 (voltages from -80 mV to -54 , -40 and -22 mV, that is

Table 2: Recovered channel number against the largest open probability reached ($N = 2000$; 400 simulated recordings per design).

$p_{\max}$	0.19	0.30	0.40	0.50	0.60	0.70	0.87	0.93
$\hat{N}$, 10%	699	1160	1270	1462	1613	1687	1770	1809
$\hat{N}$, median	1834	2105	1935	1996	2039	2010	2009	1988
$\hat{N}$, 90%	∞	11592	4904	3296	2748	2472	2240	2193
failed fits	20.5%	6.5%	1.5%	0	0	0	0	0

$p_{\max} = 0.19, 0.50, 0.87$; 400 recordings each; median $\hat{i}$ between 0.985 and 1.001 pA), because it is the slope of the parabola at the origin. The precision of $\hat{N}$ improves steadily with $p_{\max}$, and there is no threshold at the half-point. Reaching $p = \frac{1}{2}$ determines N to within about a factor of two (10–90% range 1462–3296); reaching 0.87 determines it to about $\pm 12\%$; at 0.19 there is no reliable estimate. A precise channel count needs open probabilities well above $\frac{1}{2}$ [4].

4 Voltage-dependent block

Consider a blocker of valence z_{bl} applied from the outside. It binds without permeating at electrical distance δ from the outside, with zero-voltage dissociation constant $K_{\text{d}}(0)$ (V is the inside potential relative to the outside). The site feels the fraction δ of the membrane potential, so the dissociation constant is $K_{\text{d}}(V) = K_{\text{d}}(0)e^{z_{\text{bl}}\delta FV/RT}$, and the fraction of channels left unblocked is [5]

$$B(V) = \frac{1}{1 + [\text{B}]/K_{\text{d}}(V)} = \frac{1}{1 + ([\text{B}]/K_{\text{d}}(0))e^{-z_{\text{bl}}\delta FV/RT}}.$$

Proposition 7 (the blocker as a gate). *$B(V)$ is a Boltzmann gate, $B(V) = 1/(1 + e^{-(V-V_{1/2})/k_{\text{bl}}})$, with slope $k_{\text{bl}} = RT/(z_{\text{bl}}\delta F)$ and half-point $V_{1/2} = k_{\text{bl}} \ln([\text{B}]/K_{\text{d}}(0))$. A change of affinity $K_{\text{d}}(0) \mapsto \kappa K_{\text{d}}(0)$ moves the half-point by $-k_{\text{bl}} \ln \kappa$ at fixed slope. A change of electrical distance $\delta \mapsto c\delta$ divides both k_{bl} and $V_{1/2}$ by c . Doubling the blocker concentration moves the half-point by $k_{\text{bl}} \ln 2$.*

Proof. With $k_{\text{bl}} = RT/(z_{\text{bl}}\delta F)$, $([\text{B}]/K_{\text{d}}(0))e^{-V/k_{\text{bl}}} = e^{-(V-V_{1/2})/k_{\text{bl}}}$ exactly when $V_{1/2} = k_{\text{bl}} \ln([\text{B}]/K_{\text{d}}(0))$. The three statements follow from this formula: $\ln([\text{B}]/(\kappa K_{\text{d}}(0))) = \ln([\text{B}]/K_{\text{d}}(0)) - \ln \kappa$; k_{bl} is proportional to $1/\delta$; and $\ln(2[\text{B}]) = \ln[\text{B}] + \ln 2$. $\square$

For the magnesium block of NMDA receptors [6, 7], the standard empirical form $B(V) = 1/(1 + [\text{Mg}^{2+}]/3.57 \text{ mM} \cdot e^{-0.062V/\text{mV}})$ [8] gives $k_{\text{bl}} = 1/0.062 = 16.13 \text{ mV}$ and $K_{\text{d}}(0) = 3.57 \text{ mM}$, and $V_{1/2} = k_{\text{bl}} \ln([\text{Mg}^{2+}]/3.57 \text{ mM})$ (Table 3).

Table 3: Magnesium block of NMDA receptors as a two-state gate.

$[\text{Mg}^{2+}]$ (mM)	$V_{1/2}$ (mV)	$B(-70 \text{ mV})$	$B(-20 \text{ mV})$
0.5	-31.71	0.085	0.674
1.0	-20.53	0.044	0.508
2.0	-9.35	0.023	0.341

Each doubling moves the half-point by $k_{\text{bl}} \ln 2 = 11.18 \text{ mV}$. At room temperature (295 K), $RT/F = 25.42 \text{ mV}$ (with R and F from [22]), so the slope corresponds to $z_{\text{bl}}\delta = (RT/F)/k_{\text{bl}} = 1.58$, that is $\delta \approx 0.8$ for the divalent ion. At a resting potential of -65 mV the receptor is held almost entirely in its blocked member, 44.5 mV below the half-point at 1 mM. This is

why its current requires the coincidence of glutamate binding and postsynaptic depolarisation. The two-state form describes an impermeant blocker. Magnesium also permeates the channel at strongly negative voltages [23], which adds a second exit from the blocked state and relieves the block there.

5 The spike threshold

Let Φ , m_∞ , V_a , k_a and the threshold θ , the voltage where $\Phi'(\theta) = 0$, be as in Theorem 4, following Platkiewicz and Brette [10] (see also [9]). Since $m'_\infty = m_\infty(1 - m_\infty)/k_a$,

$$\Phi'(V) = -g_L + g_{\text{Na}}m_\infty(V) \left[\frac{(E_{\text{Na}} - V)(1 - m_\infty(V))}{k_a} - 1 \right].$$

Below V_a , $m_\infty \approx e^{(V-V_a)/k_a}$ and $1 - m_\infty \approx 1$. In this exponential approximation $\Phi'(\theta) = 0$ becomes $g_{\text{Na}}e^{(\theta-V_a)/k_a}[(E_{\text{Na}} - \theta)/k_a - 1] = g_L$.

We prove Theorem 4(a).

Proof. Taking logarithms of $g_{\text{Na}}e^{(\theta-V_a)/k_a}[(E_{\text{Na}} - \theta)/k_a - 1] = g_L$ gives the fixed-point equation. Substitute $y = (E_{\text{Na}} - \theta)/k_a - 1$, so that $\theta = E_{\text{Na}} - k_a(1+y)$ and $(\theta - V_a)/k_a = (E_{\text{Na}} - V_a)/k_a - 1 - y$. The equation $e^{(\theta-V_a)/k_a}y = g_L/g_{\text{Na}}$ becomes $ye^{-y} = (g_L/g_{\text{Na}})e^{1-(E_{\text{Na}}-V_a)/k_a}$, that is $(-y)e^{-y} = w$ with $w = -(g_L/g_{\text{Na}})e^{1-(E_{\text{Na}}-V_a)/k_a}$. For $w \in (-1/e, 0)$ the equation $Ue^U = w$ has exactly two real solutions, $U = W_0(w) \in (-1, 0)$ and $U = W_{-1}(w) < -1$ [24]. Hence $y = -W_0(w) \in (0, 1)$ or $y = -W_{-1}(w) > 1$. The first gives $\theta > E_{\text{Na}} - 2k_a$. The second gives $\theta < E_{\text{Na}} - 2k_a$ and the stated formula. $\square$

For the parameters below, w lies between -10^{-6} and -10^{-7} . The branch W_0 gives $\theta = 49.0$ mV, far above V_a , where the exponential approximation does not hold; the branch W_{-1} gives the threshold. Fixed points and bifurcations of exponential integrate-and-fire models are analysed in [25]. With an exponential nonlinearity such fixed-point conditions are of Lambert type, and the form here carries the sodium driving force. Replacing $E_{\text{Na}} - \theta$ by $E_{\text{Na}} - V_a$ and dropping the -1 gives the simplest form of the threshold equation of [10], $\theta \approx V_a - k_a \ln[(g_{\text{Na}}/g_L)(E_{\text{Na}} - V_a)/k_a]$.

We take $V_a = -30$ mV and $k_a = 6$ mV, within the ranges quoted in [10] (V_a about -30 mV, $k_a = 4$ – 8 mV for neuronal sodium channels), and the illustrative values $E_{\text{Na}} = 55$ mV and $E_L = -70$ mV. The exact θ is the root of the full $\Phi'(V) = 0$ below V_a (with the Boltzmann m_∞), found by a bracketing root search on $[-150, V_a]$ mV. For $V < \theta$, $\Phi' < 0$, so Φ decreases from $+\infty$ to its minimum $\Phi(\theta)$, and a resting state ($\Phi = 0$ below θ) exists if and only if $\Phi(\theta) < 0$. Solving $\Phi(\theta) = 0$ as a function of g_{Na}/g_L shows that the model has a resting state exactly for $g_{\text{Na}}/g_L < 14.65$; above that, Φ has no zero below θ and θ is the minimum of Φ . Table 4 covers the range with a resting state (errors are approximation minus exact).

Table 4: Spike threshold: exact root, simplest threshold equation and closed form.

g_{Na}/g_L	2	4	6	8	10	12	14
exact θ (mV)	-50.63	-55.27	-57.92	-59.78	-61.21	-62.38	-63.37
simplest form, error (mV)	+0.56	+1.04	+1.26	+1.39	+1.49	+1.57	+1.63
closed form, error (mV)	-0.42	-0.19	-0.12	-0.09	-0.07	-0.06	-0.05

The closed form reduces the error of the simplest form by a factor of 1.3 (at $g_{\text{Na}}/g_L = 2$) to 33 (at 14). Its remaining error is that of the exponential approximation to m_∞ , and it shrinks as θ moves below V_a .

We prove Theorem 4(b), which states what a threshold reading determines.

Proof. $g_{\text{Na}}e^{(V-V_a)/k_a} = (g_{\text{Na}}e^{-V_a/k_a})e^{V/k_a}$, which is unchanged when V_a moves by Δ and g_{Na} is multiplied by e^{Δ/k_a} . In part (a) the argument of W_{-1} is $-e^{1-E_{\text{Na}}/k_a} e^{(V_a-k_a \ln(g_{\text{Na}}/g_L))/k_a}$. $\square$

6 The spike as a pair: the 3 + 1 sodium channel and the 3:2 pump

Background. Hodgkin and Huxley described the sodium conductance as $g_{\text{Na}}m^3h$: three identical activation particles and one inactivation particle [1]. The voltage-gated sodium channel is one polypeptide with four homologous domains, I–IV, each with a voltage sensor [17]; its primary voltage sensor comprises four positively charged S4 segments [11]. Fluorescence near the S4 segments of domains I, II and III has a fast component comparable to the fast gating current, and the three signals start together. The domain IV signal follows the slow gating component with a lag, a later step in the activation sequence [11]. Movement of the domain IV sensor is the rate-limiting step for the development of and recovery from fast inactivation, and its activation alone is sufficient for fast inactivation [12]. The overlap of inward and outward currents is the largest component of the energetic cost of the squid-axon spike in Hodgkin–Huxley-type models [26], and sodium entry relative to its capacitive minimum is a standard measure of spike efficiency [27].

We prove Theorem 5(a).

Proof. Integrate the current equation: $\int C\dot{V} dt = C\Delta V$, which is 0 in the first case and $C\Delta V_{\text{sp}}$ in the second. $\square$

$C\Delta V_{\text{sp}}$ is the charge the membrane capacitance needs to reach the peak. The ratio $\rho = Q_{\text{Na}}/(C\Delta V_{\text{sp}})$ compares the sodium charge with it, and $Q_{\text{Na}} - C\Delta V_{\text{sp}}$ is the sodium charge cancelled by outward current.

The squid-axon spike. The membrane is

$$C\dot{V} = -g_{\text{Na}}m^3h(V - E_{\text{Na}}) - g_{\text{K}}n^4(V - E_{\text{K}}) - g_{\text{L}}(V - E_{\text{L}}) + I_{\text{stim}},$$

with $\dot{x} = \alpha_x(V)(1 - x) - \beta_x(V)x$ for $x = m, h, n$ and the rate functions (1). The standard squid parameters are $C = 1 \mu\text{F}/\text{cm}^2$, $g_{\text{Na}} = 120$, $g_{\text{K}} = 36$, $g_{\text{L}} = 0.3 \text{ mS}/\text{cm}^2$, $E_{\text{Na}} = 50$, $E_{\text{K}} = -77$ and $E_{\text{L}} = -54.387 \text{ mV}$ [1, 21]. The equations, together with the cumulative integrals of the three ionic currents and of I_{stim} , are integrated numerically by an adaptive stiff/non-stiff multistep method (relative tolerance 10^{-9} , absolute 10^{-11} , step at most 0.01 ms), and the solution is read on a $1 \mu\text{s}$ grid. An independent integration by an implicit Runge–Kutta (Radau) method reproduces the m^3h values below to the digits quoted. The resting state is the fixed point of the four-dimensional system at $I_{\text{stim}} = 0$, found as the root of the steady-state current and confirmed stable by the eigenvalues of the Jacobian: the membrane rests at -64.996 mV . The stimulus starts at $t = 1 \text{ ms}$. The amplitude is $\Delta V_{\text{sp}} = V_{\text{peak}} - V_{\text{rest}}$, and the width is the full width at $V_{\text{rest}} + \Delta V_{\text{sp}}/2$. Charges are integrated from stimulus onset to the first return of V to V_{rest} after the peak. The 1-ms threshold is the smallest pulse amplitude for which V exceeds 0 mV within 10 ms of onset, found by bisection.

A pulse of $10 \mu\text{A}/\text{cm}^2$ for 1 ms (threshold $6.919 \mu\text{A}/\text{cm}^2$) gives a spike peaking at 39.067 mV : amplitude 104.063 mV , width at half amplitude 1.4544 ms (Fig. 2c). From onset to repolarisation the membrane takes in $Q_{\text{Na}} = 1389.934 \text{ nC}/\text{cm}^2$ of sodium charge against $C\Delta V_{\text{sp}} = 104.063 \text{ nC}/\text{cm}^2$: $\rho = 13.3566$ (13.4663 over a 20 ms window). The rising phase alone uses $1.6901 C\Delta V_{\text{sp}}$, and 87.35% of the sodium charge enters after the peak. At the peak of the inward sodium current ($787.6 \mu\text{A}/\text{cm}^2$, at 2.911 mV) the potassium current is $821.2 \mu\text{A}/\text{cm}^2$, and the net ionic

current is 0.0645 of the sodium current. Theorem 5(a), evaluated with the integrated charges, holds to 1.0×10^{-16} of Q_{Na} , that is to rounding error.

The number of activating particles. We replace m^3 by m^q in the sodium current, with all other parameters fixed (Table 5, Fig. 2d). Fixed points are the roots of the steady-state current on $[-100, 50]$ mV, bracketed on a 0.005-mV grid and refined by Brent’s method, with stability from the eigenvalues of the Jacobian. For $q = 2$ the limit cycle is obtained by integrating for 400 ms from 1 mV above the fixed point. Its period is averaged over the cycles in the last 200 ms, and ρ per cycle is the sodium charge per cycle divided by C times the peak-to-trough amplitude.

Table 5: The sodium conductance $m^q h$ for $q = 1, \dots, 4$, other parameters fixed.

q	fixed point (mV)	standard stimulus	amplitude (mV)	width (ms)	ρ
1	−40.32, stable	no spike	—	—	—
2	−47.43, unstable	periodic firing	115.346 ^a	1.60	14.0119 ^a
3	−65.00, stable	spike	104.063	1.4544	13.3566
4	−65.83, stable	no spike	104.533 ^b	1.4635 ^b	11.9407 ^b

^aPeak to trough and per cycle of the limit cycle. ^bAt 1.5 times the $m^4 h$ 1-ms threshold of 22.326 $\mu\text{A}/\text{cm}^2$, that is at 33.489 $\mu\text{A}/\text{cm}^2$.

With one activating particle the steady window current holds the membrane at −40.32 mV with the inactivation gate almost closed ($h = 0.0522$). The standard pulse moves it by 1.072 mV, and it crosses 0 mV only when driven by a pulse 70.99 times the $m^3 h$ threshold: there is no regenerative spike. With two particles no resting state is stable, and the membrane fires spontaneously with period 10.4268 ms. With four the rest is stable, and the standard pulse gives only 7.363 mV because the threshold is 3.227 times higher; above threshold the spike is almost unchanged. Among the spiking cases ρ is 14.0119 ($q = 2$, per cycle), 13.3566 ($q = 3$) and 11.9407 ($q = 4$). Within the fitted model of [1], whose other parameters were fitted with three activating particles, $q = 3$ is the only exponent that gives both a stable rest and a spike to the standard stimulus. This proves Theorem 5(b).

The pump as the unequal pair 3:2. The sodium–potassium pump exports three Na^+ and imports two K^+ per ATP hydrolysed [19]. We prove Theorem 6.

Proof. As a pair (3, 2) the stoichiometry has common part $\frac{5}{2}$ and half-difference $\pm\frac{1}{2}$, and the net charge moved outwards per cycle, $3 - 2 = 1$ elementary charge, is twice the half-difference. Moving one Na^+ from inside to outside against its electrochemical gradient costs $e(E_{\text{Na}} - V_m)$, and moving one K^+ from outside to inside costs $e(V_m - E_{\text{K}})$. Summing over the three and the two ions gives $W_{\text{cyc}} = e(3E_{\text{Na}} - 2E_{\text{K}} - V_m)$, whose derivative in V_m is $-e$: one electron-volt per volt, because one net charge is moved outwards. The cycle runs forward when the free energy supplied exceeds the work against the gradients, $|\Delta G_{\text{ATP}}| > W_{\text{cyc}}$, that is when $V_m > 3E_{\text{Na}} - 2E_{\text{K}} - |\Delta G_{\text{ATP}}|/e$; at equality it stalls, and below it the net cycle reverses. $\square$

With the illustrative values $E_{\text{Na}} = +55$ mV, $E_{\text{K}} = -90$ mV and $V_m = -65$ mV, and potentials in volts so that the costs are in eV, the work against the gradients per cycle is

$$W_{\text{cyc}} = 3(E_{\text{Na}} - V_m) + 2(V_m - E_{\text{K}}) = 0.360 + 0.050 = 0.410 \text{ eV} = 15.3 k_B T \text{ at } 310 \text{ K}.$$

A 2:2 exchanger would do $2 \cdot 0.120 + 2 \cdot 0.025 = 0.290$ eV and move no charge. With $|\Delta G_{\text{ATP}}| \approx 0.55$ eV per ATP (53 kJ mol^{−1}, within the range of about 47–70 kJ mol^{−1} for cells collected in [20]), the pump stalls where $W_{\text{cyc}} = |\Delta G_{\text{ATP}}|$, at $V_m = 3E_{\text{Na}} - 2E_{\text{K}} - 0.55 \text{ V} = -205$ mV,

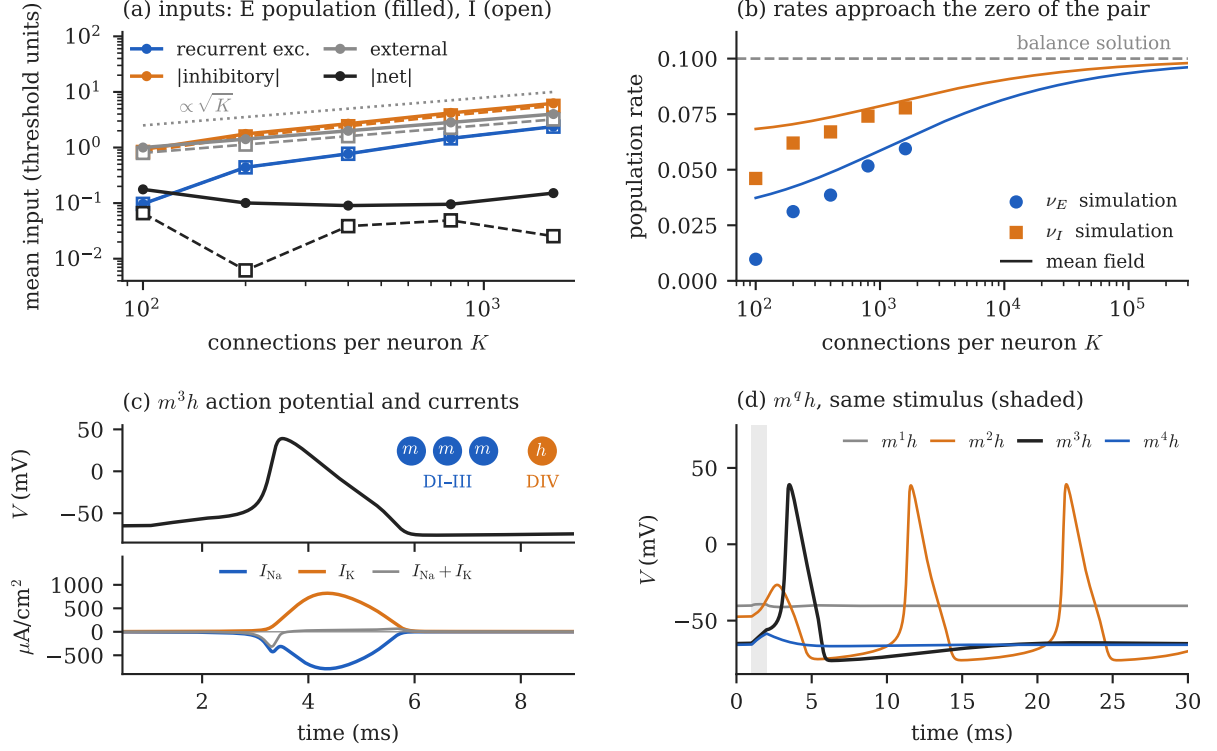


Figure 2: (a) Mean inputs against K ; E population filled, I open (recurrent excitation is the same for both); the dotted line has slope $\frac{1}{2}$ ($\propto \sqrt{K}$). External input grows as $\sqrt{K}$ by construction; excitation and inhibition grow faster here because the rates are still rising; the net input stays of order one. (b) Population rates ν_E , ν_I and the balance solution 0.1; lines: mean-field theory. (c) The $m^3 h$ action potential (Hodgkin–Huxley squid parameters, $6.3^\circ C$) with its sodium and potassium currents and their sum; inset: three activating particles (domains I–III), one inactivating (domain IV). (d) $m^q h$, $q = 1, \dots, 4$, other parameters fixed, same $10 \mu A/cm^2$, 1 ms stimulus.

and reverses its cycle at more negative potentials. It therefore operates forward throughout the physiological range. Exporting the sodium that enters during one squid-axon spike takes $Q_{Na}/(3e) = 2.89 \times 10^{12}$ pump cycles, and as many ATP molecules, per cm^2 of membrane.

7 The balanced cortex as the zero of a pair

Background. Van Vreeswijk and Sompolinsky [13, 14] studied networks of N_E excitatory and N_I inhibitory binary neurons. Each neuron receives on average $K \ll N_E, N_I$ connections of strength $O(1/\sqrt{K})$ from each population, and an external excitatory input of order $\sqrt{K}$. The excitatory and inhibitory inputs are then each of order $\sqrt{K}$, far above threshold, yet a balance between them emerges dynamically for a wide range of parameters. The large inhibitory input nearly cancels the large excitatory one, the net input is of order threshold, and the network is in an asynchronous chaotic state with irregular, fluctuation-driven firing. To leading order the rates solve linear balance equations.

Model. Neuron j of population $X \in \{E, I\}$ has state $S_j \in \{0, 1\}$ and input

$$u_j = \frac{1}{\sqrt{K}} \left(\sum_{l \in E} c_{jl} S_l - J_X \sum_{l \in I} c_{jl} S_l \right) + \sqrt{K} b_X \nu_0, \quad S_j \leftarrow \Theta(u_j - \theta_X),$$

with Θ the unit step, $c_{jl} = 1$ independently with probability K/N_Y for l in population Y (and 0 otherwise), $J_E = 2$, $J_I = 1.8$, $b_E = 1$, $b_I = 0.8$, $\nu_0 = 0.1$, $\theta_E = 1$ and $\theta_I = 0.7$. The population

rates ν_E, ν_I are the fractions of active neurons, and ν, μ, A and b are as in Section 1.1.

Proposition 8 (the triad of weights). *Every row of the connectivity matrix takes values in $\{+1, 0, -J_X\}/\sqrt{K}$. Averaged over the connectivity, the mean excitatory and inhibitory inputs to population X are $\sqrt{K} \nu_E$ and $-\sqrt{K} J_X \nu_I$, up to $O(\sqrt{K}/N_E)$ terms from the excluded self-connections. The population-mean net input is therefore $\mu = \sqrt{K} (b \nu_0 + A \nu) + O(\sqrt{K}/N_E)$.*

Proof. An entry of row j is 0 when $c_{jl} = 0$, $+1/\sqrt{K}$ for an excitatory l with $c_{jl} = 1$, and $-J_X/\sqrt{K}$ for an inhibitory one. Each presynaptic neuron adds K/N_Y in expectation to each postsynaptic count, so summing over population Y gives K times its rate, less KS_j/N_Y for a self-connection. Dividing by $\sqrt{K}$ and adding the external input gives μ . $\square$

We prove Theorem 1.

Proof. The balance solution is defined by $b \nu_0 + A \nu_{\text{bal}} = 0$, that is $\nu_E - J_E \nu_I = -b_E \nu_0$ and $\nu_E - J_I \nu_I = -b_I \nu_0$. Subtracting the second equation from the first gives $(J_E - J_I) \nu_I = (b_E - b_I) \nu_0$, and then $\nu_E = J_E \nu_I - b_E \nu_0 = \nu_0 (J_I b_E - J_E b_I) / (J_E - J_I)$. Here $\det A = J_E - J_I$ and

$$A^{-1} = \frac{1}{J_E - J_I} \begin{pmatrix} -J_I & J_E \\ -1 & 1 \end{pmatrix}.$$

(a) All parameters are positive. If $J_E > J_I$, positivity needs $b_E > b_I$ and $J_I b_E > J_E b_I$, and the second implies the first; the case $J_E < J_I$ reverses both inequalities. (b) $b \nu_0 = -A \nu_{\text{bal}}$; insert this in Theorem 8. (c) is (b) read with $\mu = O(1)$. $\square$

Our parameters give $b_E/b_I = 1.2500 > J_E/J_I = 1.1111 > 1$, $\nu_E^{\text{bal}} = \nu_I^{\text{bal}} = 0.1$, $\det A = 0.2$ and

$$A^{-1} = \begin{pmatrix} -9 & 10 \\ -5 & 5 \end{pmatrix}.$$

Each row is a ternary weight vector in the sense of [18]. Its expected fraction of zero entries is $1 - K/N_Y$ in each population block, from 0.99 at $K = 100$ to 0.84 at $K = 1600$ for the networks below.

Simulation method. $N_E = N_I = 10^4$ and $K \in \{100, 200, 400, 800, 1600\}$, so K/N_E ranges from 0.01 to 0.16 (the largest network has 63,992,591 synapses). Each connection c_{jl} , $l \neq j$, is drawn once, independently, with probability K/N_Y , and each neuron starts active with probability ν_0 . Each E (I) neuron updates at Poisson times of rate $1/\tau_E$ ($1/\tau_I$), with $\tau_E = 1$ (about 10 ms) and $\tau_I = 0.9$. This is approximated by steps $dt = 0.01 \tau_E$ in which each neuron updates independently with probability dt/τ_X , using the inputs at the start of the step. The synaptic counts $\sum_{l \in E} c_{jl} S_l$ and $\sum_{l \in I} c_{jl} S_l$ are integers; they are updated at every change of state and agree exactly with a full recount at the end of each simulation. We discard $10 \tau_E$ and measure over $40 \tau_E$, sampling every $0.1 \tau_E$ (400 samples). Connectivity, initial state and update times come from one fixed pseudo-random sequence for each K . The population rate ν_X is the fraction of active neurons in population X , averaged over the samples. An input component is its value averaged over the neurons of a population and over the samples. A log-log slope is the least-squares slope of $\log |\cdot|$ against $\log K$ over the five values of K . With $dt = 0.002$ the rates at $K = 1600$ are 0.05922 and 0.07774 (against 0.05944 and 0.07782).

(a) The components and the zero. At $K = 1600$ (Table 6, Fig. 2a) the E population receives 2.3768 (recurrent excitation) $+ 4.0000$ (external) $- 6.2254$ (inhibition) $= 0.1514$, and the I population $2.3770 + 3.2000 - 5.6024 = -0.0254$. The net input is thus a fraction 0.0237 (E) and 0.0046 (I) of the excitatory drive: fractions 0.9763 and 0.9954 of it cancel. The net input has log-log slopes -0.0523 (E) and 0.0234 (I), so it stays of order one. The excitatory and inhibitory inputs grow as $\sqrt{K}$ per unit rate: the per-unit-rate components exc/ν_E and $|\text{inh}|/(J_X \nu_I)$ have log-log slopes 0.4999 and 0.5003 (E) and 0.5004 and 0.4994 (I), and divided by $\sqrt{K}$ they lie between 0.9986 and 1.0020 at every K . The raw slopes of the components, 1.0962 and 1.0966 (excitation) and 0.6774 and 0.6765 (inhibition), reflect the still-rising rates: ν_E grows with slope 0.5963 and ν_I with 0.1771 , and by Theorem 8 a component's slope is $\frac{1}{2}$ plus the slope of its rate.

Table 6: Mean inputs (threshold units) and rates; relative error $(\nu - \nu_{\text{bal}})/\nu_{\text{bal}}$. Mean field (MF, large N_E, N_I [14]): $\nu_X = H((\theta_X - \mu_X)/\sqrt{\Sigma_X})$, with μ from Theorem 8, $\Sigma_X = \nu_E + J_X^2 \nu_I$ and $H(x) = \frac{1}{2} \text{erfc}(x/\sqrt{2})$ the Gaussian tail. The two equations are solved for (ν_E, ν_I) by Newton-type root finding started at ν_{bal} , to a residual below 10^{-13} .

K	exc.	inh. (E)	net (E)	net (I)	ν_E (rel. err.)	ν_I (rel. err.)	MF ν_E	MF ν_I
100	0.0969	-0.9202	0.1767	0.0660	0.00970 (-0.9030)	0.04606 (-0.5394)	0.03731	0.06834
200	0.4408	-1.7544	0.1006	-0.0062	0.03115 (-0.6885)	0.06200 (-0.3800)	0.04224	0.07050
400	0.7714	-2.6811	0.0903	-0.0385	0.03858 (-0.6142)	0.06701 (-0.3299)	0.04866	0.07361
800	1.4613	-4.1943	0.0954	-0.0489	0.05169 (-0.4831)	0.07411 (-0.2589)	0.05615	0.07737
1600	2.3768	-6.2254	0.1514	-0.0254	0.05944 (-0.4056)	0.07782 (-0.2218)	0.06398	0.08137

(b) Rates and the balance solution. The rates approach $\nu_{\text{bal}} = 0.1$ monotonically and slowly (Table 6, Fig. 2b): at $K = 1600$ they are 40.6% (E) and 22.2% (I) below it, exactly as Theorem 1(b) requires. Inserting the measured population-mean net inputs μ into $\nu = \nu_{\text{bal}} + A^{-1}\mu/\sqrt{K}$ reproduces the measured rates to within 2.67×10^{-3} at every K (1.45×10^{-4} at $K = 1600$); the remainder is the quenched spread of out-degrees. The residual is of order one, and A^{-1} multiplies it by up to 10. At $K = 1600$ the simulated rates lie 7.10% (E) and 4.35% (I) below the mean-field values. In one further simulation with $N_E = N_I = 2 \times 10^4$ ($K/N_E = 0.08$) they move towards them, to 0.06110 and 0.07927 , so this gap is a finite- K/N_E effect. In mean field the rate error falls below 10% only for $K > 40,054$ (E) and $K > 8,897$ (I), the roots in K of $|\nu_X - 0.1| = 0.01$. As $K \rightarrow \infty$, $\nu_X \rightarrow 0.1$, so the mean-field equation forces $(\theta_X - \mu_X)/\sqrt{\Sigma_X} \rightarrow H^{-1}(0.1) = 1.28155$ with $\Sigma_X = 0.1(1 + J_X^2)$, that is $\mu_E \rightarrow 0.0938$ and $\mu_I \rightarrow -0.1345$. By Theorem 1(b), $\sqrt{K}$ times the relative error then tends to $A^{-1}\mu_\infty/0.1 = (-21.891, -11.415)$ for (E, I). Over our range the slopes of $\log|\nu - \nu_{\text{bal}}|$, -0.2821 (E) and -0.3118 (I), are pre-asymptotic.

(c) Irregularity: firing driven by the residual. At $K = 1600$ the net input fluctuates in time with standard deviation 0.5108 (E) and 0.4691 (I) (the standard deviation of each neuron's net input over the 400 samples, averaged over the population). Its mean lies 1.6613 (E) and 1.5462 (I) standard deviations below threshold, so the neurons fire only on fluctuations: the fraction of time the net input exceeds threshold is 0.06032 (E) and 0.07784 (I), close to the rates 0.05944 and 0.07782 . Time-averaged single-neuron activities are broadly spread: mean 0.0594 , standard deviation 0.0702 (coefficient of variation 1.182) in E and 0.0778 , 0.0800 (coefficient of variation 1.028) in I, with 32.21% of E and 22.53% of I neurons below 0.01 . The single-neuron autocorrelation decays with $1/e$ time $1.171 \tau_E$ (E) and $1.104 \tau_E$ (I), close to the update time. It is computed from the activity fluctuations of 1000 recorded neurons per population, averaged over neurons, at lags that are multiples of $0.1 \tau_E$, and the $1/e$ time is found by linear interpolation. The population activity fluctuates over the samples with standard deviation 0.00290 (E) and

0.00164 (I), against $\sqrt{\nu_X(1-\nu_X)/N_X} = 0.00236$ and 0.00268 for independent neurons. The two are of the same order: the state is asynchronous.

8 Discussion: the pair, the triad and the fourth across the scales

The pair and its zero. At the gate, the two members are balanced at the half-point, where the log-odds is zero, the occupancy $\frac{1}{2}$, the entropy one bit and the fluctuation largest (Theorem 2). A membrane reading determines a gate's half-point and slope from its mean. It determines the unitary current from fluctuations at any open probability, and the number of channels increasingly well the further the gate is driven beyond its half-point (Section 3). It separates a blocker's affinity from its electrical distance through the two parameters of one Boltzmann pair (Theorem 7). It determines the spike threshold only through $g_{\text{Na}}e^{-V_a/k_a}$ (Theorem 4(b)). In the network, excitation and inhibition are s and $-s$. Each is of order $\sqrt{K}$, exactly $\sqrt{K}$ per unit rate, and their sum with the external drive is of order one. The balance solution is the rate vector at which the zero is exact at order $\sqrt{K}$.

The merge. Adding independent identical gates keeps the pair: the normalised mean current is $p(V)$ with the half-point of one gate, and the variance of the sum is largest at the same half-point (Theorem 3). Multiplying gates does not keep it (n_∞^4 , Section 2).

The triad. Each connectivity row is ternary, $\{+1, 0, -J_X\}/\sqrt{K}$ (Theorem 8), and the zero of the network is built from these triads. The same three-valued alphabet carries the ternary-weight networks and balanced-ternary arithmetic of [18].

The residual. What is not cancelled drives the output. Neurons fire when the order-one net input fluctuates above threshold, and Theorem 1 shows that the same residual sets the distance from the zero, $\nu - \nu_{\text{bal}} = A^{-1}\mu/\sqrt{K}$. In our range of K the inputs are balanced while the rates are still approaching the balance solution; $\det A = J_E - J_I$ sets how far K must grow. At the spike, the inward sodium charge equals the outward potassium and leak charge less the stimulus charge, exactly, over any interval that returns to the starting voltage (Theorem 5(a)). The voltage excursion is the part of the sodium charge not yet cancelled, and the cost of the spike, $\rho = 13.36$, is the overlap of the two members. The pump restores the gradients as the unequal pair 3:2, whose half-difference is the one net charge it moves per cycle.

The fourth element. In m^3h three identical members open the channel and a fourth closes it. In the channel, three sensors move together and fast, and the fourth moves later and is, on its own, sufficient and rate limiting for fast inactivation [11, 12]: it closes what the other three open. In the network, each neuron reads a zero-sum ternary input, and its output becomes an entry of other neurons' ternary rows: the observer of one triad is a member of the next.

Outlook. Theorem 1 turns a recording of excitatory and inhibitory currents into a reading of how far a circuit sits from its balance solution, and the next step is to apply it to spiking networks with conductance-based synapses and to recorded cortical currents. For channel counting, recordings that drive the gate well beyond its half-point give the largest gain in precision.

9 Data and sources

No experimental data set is used. Every number is computed from the following inputs by the formulas and procedures stated in the text.

- *Hodgkin-Huxley rate functions:* the fits of Hodgkin and Huxley for the squid giant axon at 6.3°C [1], in the form shifted to a resting potential of -65 mV given in [21, Ch. 5], as written in (1); used in Sections 2 and 6.

- *Hodgkin–Huxley membrane parameters* [1, 21]: $C = 1 \mu\text{F}/\text{cm}^2$; $g_{\text{Na}} = 120$, $g_{\text{K}} = 36$, $g_{\text{L}} = 0.3 \text{ mS}/\text{cm}^2$; $E_{\text{Na}} = 50$, $E_{\text{K}} = -77$, $E_{\text{L}} = -54.387 \text{ mV}$. The stimulus ($10 \mu\text{A}/\text{cm}^2$ for 1 ms) is a chosen value.
- *Magnesium block*: the empirical form $B(V) = 1/(1 + [\text{Mg}^{2+}]/3.57 \text{ mM} \cdot e^{-0.062V/\text{mV}})$ of Jahr and Stevens [8], that is $K_{\text{d}}(0) = 3.57 \text{ mM}$ and $k_{\text{bl}} = 1/0.062 \text{ mV}$; the concentrations 0.5, 1 and 2 mM and the temperature 295 K used for RT/F are chosen values.
- *Physical constants* (CODATA 2018 [22], all exact in the SI): $k_{\text{B}} = 8.617333262 \times 10^{-5} \text{ eV K}^{-1}$, $R = 8.314462618 \text{ J mol}^{-1} \text{ K}^{-1}$, $F = 96485.33212 \text{ C mol}^{-1}$, $e = 1.602176634 \times 10^{-19} \text{ C}$.
- *Threshold model*: $V_{\text{a}} = -30 \text{ mV}$ and $k_{\text{a}} = 6 \text{ mV}$, within the ranges quoted in [10]; $E_{\text{Na}} = 55 \text{ mV}$ and $E_{\text{L}} = -70 \text{ mV}$ are illustrative values; the ratios $g_{\text{Na}}/g_{\text{L}}$ are chosen.
- *Sodium–potassium pump*: stoichiometry 3 Na^+ out, 2 K^+ in per ATP [19]; $E_{\text{Na}} = +55 \text{ mV}$, $E_{\text{K}} = -90 \text{ mV}$, $V_{\text{m}} = -65 \text{ mV}$ and the temperature 310 K are illustrative values; $|\Delta G_{\text{ATP}}| \approx 0.55 \text{ eV}$ per molecule, within the cellular range collected in [20].
- *Simulated recordings* (Section 3): model values $N = 2000$, $i = 1 \text{ pA}$, $\sigma_b = 3 \text{ pA}$, gate $V_{1/2} = -40 \text{ mV}$ and $k = 9.47 \text{ mV}$ (the m gate of Table 1, rounded), 9 voltages per recording, 200 sweeps per voltage and 400 recordings per design, all chosen. The recordings are pseudo-random samples generated as described in the text and contain no external input.
- *Network parameters* (Section 7): the binary-network model of [13, 14]. Its parameter values ($N_{\text{E}} = N_{\text{I}} = 10^4$, $K = 100\text{--}1600$, $J_{\text{E}} = 2$, $J_{\text{I}} = 1.8$, $b_{\text{E}} = 1$, $b_{\text{I}} = 0.8$, $\nu_0 = 0.1$, $\theta_{\text{E}} = 1$, $\theta_{\text{I}} = 0.7$, $\tau_{\text{E}} = 1$, $\tau_{\text{I}} = 0.9$, $\text{dt} = 0.01 \tau_{\text{E}}$) are chosen values. $H^{-1}(0.1) = 1.28155$ is the upper 10% point of the standard normal distribution.

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